Introduction to Radiobiological Models

Souvik Das

Resisdent Medical Physicist
Tata Memorial Hospital, Mumbai

Table of Contents

Table of Contents

Target theory

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- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

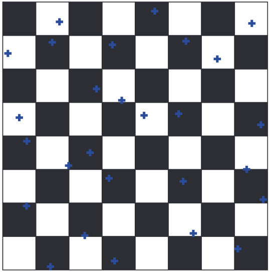

Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

Mathematical Modeling

Poisson Statistics

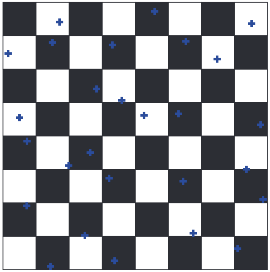

Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

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- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

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Note

D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

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Note

To achieve a certain TCP, you should aim for a tumor cell survival of (1 - TCP).

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.

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  where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.
- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

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  where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.
- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

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Multitarget, Single-Hit Model

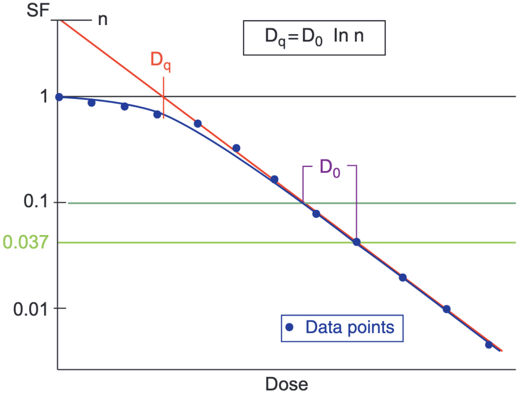

Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

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  - D_q tells you how wide the shoulder is.
  - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

Single-Hit, Multitarget Model

Advantages and Disadvantages

Fractionated Radiation and Effective D_0

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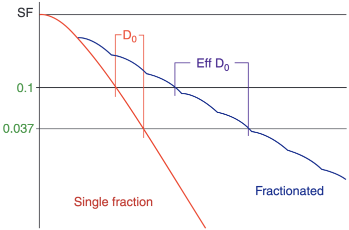

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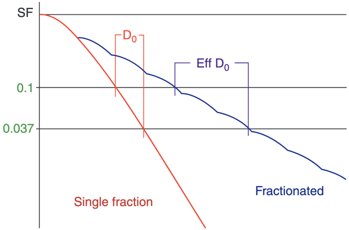
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- D_0 versus effective D_0. When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective D_0 is always larger than D_0



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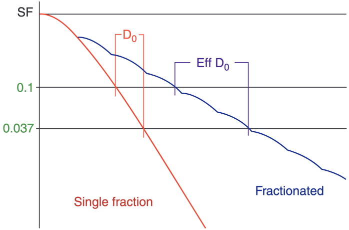

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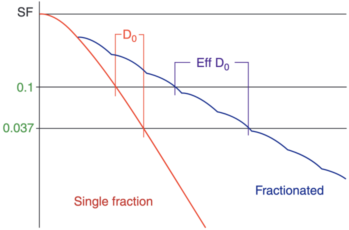

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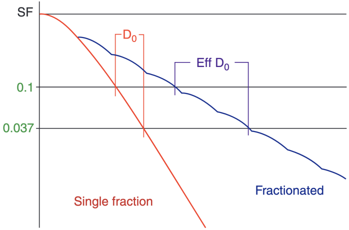
- After x fractions to total dose xD Gy: $SF_x^D = e^{-xD/\text{effective } D_0}$



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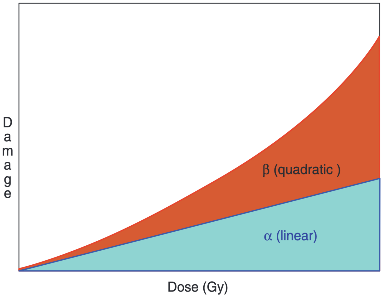
  - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.
  - Multi-hit kill (β) is repairable damage and is **dependent** on fractionation or dose rate. This corresponds to double-hit and **inter**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

Continue:

- The α/β ratio is the **dose** at which α kill and β kill are equal.
  - Low α/β ratio ("high repair") tissues are relatively resistant at small fraction size and relatively sensitive at large fraction size.
  - High α/β ratio ("low repair") tissues are relatively sensitive at small fraction size and relatively resistant at large fraction size.



Linear-Quadratic (LQ, Alpha-Beta) Model

Alpha-Beta Model and Dose Fractionation

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$\mathrm{SF}_n = e^{-n(\alpha d + \beta d^2)}$$

$$\implies -\frac{1}{nd} = \frac{\alpha}{\ln(\mathrm{SF}_n)} + \frac{\beta d}{ln(\mathrm{SF}_n)}$$

- Plot of dose per fraction d versus $-\frac{1}{nd}$ gives a straight line with slope $\beta / ln(\mathrm{SF}_n)$ and y-intercept $ln(\mathrm{SF}_n)$. The ratio of intercept to slope gives α/β.
- Such values are obtained for a variety of tissues and tumors, and are used to guide clinical decision-making regarding fractionation schemes.

- Most acute-reacting tissues are believed to have an α/β ratio $\approx$ 10.
- Most late-reacting tissues are believed to have an α/β ratio $\approx$ 3 or less.
- CNS tissue (brain, cord) is believed to have an α/β ratio $\approx$ 1-2.
- Most tumors are believed to have an α/β ratio $\geq$ 10.
- Some low α/β tumors may have an α/β ratio $\approx$ 1.5-4.Breast and Prostate are the "classic" low α/β tumors.

- One advantage of the linear-quadratic model is that it is easy to calculate effective doses for different fraction sizes.

Definition

Biologically effective dose (BED) is an extrapolated dose given over infinitely many fractions.

- For n fractions of d dose per fraction:

$$\begin{aligned}\mathrm{BED}_{\alpha/\beta} &= nd\left(1 + \frac{d}{\alpha/\beta}\right)\\ &= D\left(1 + \frac{d}{\alpha/\beta}\right)\end{aligned}$$

- BED $=$ total physical dose $\times$ relative effectiveness of that dose.

Problem 1

If a treatment is planned for 70 Gy/ 35 fractions, but due to departmental contraints, we need to complete it in 25 fractions. What is the new dose per fraction for same tumor control with $\alpha/\beta = 10$?

- EQD_2 is the dose in 2 Gy fractions that would produce the same biological effect as the actual treatment.
- For n fractions of d dose per fraction:

$$\mathrm{EQD}_2 = \frac{BED}{1 + \frac{2}{\alpha/\beta}}$$

$$\implies \mathrm{EQD}_2 = D\left(\frac{d + \alpha/\beta}{2 + \alpha/\beta}\right)$$

Problem 2

You want to treat a lung cancer to 60 Gy in 2 Gy fractions, but the first five fractions are given at 3 Gy/fx due to SVC syndrome. Assuming α/β $= 3$, how much additional dose should you deliver at 2 Gy per fraction?

MCQ Example

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

1. Total Dose
2. Dose per fraction
3. BED
4. Relative effetiveness of that dose

MCQ Example

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

3. **BED**

MCQ Example

Question 2: A tumour containing 10^8 cells is inactivated by radiation with a mean lethal dose of $D_0 = 1.8$ Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

1. 1.2×10^{-3}
2. 1.4×10^{-4}
3. 1.4×10^{-3}
4. 1.2×10^{-4}

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Solution

The number of surviving cells is given by:

$$N = N_0 e^{-D/D_0}$$

where N_0 is the initial number of cells, D is the dose, and D_0 is the mean lethal dose. Substituting the values, we get:

$$N = 10^8 e^{-45/1.8} \approx 1.4 \times 10^{-3}$$

MCQ Example

Question 3: It is required to replace an LDR implant of 60 Gy at 0.37 Gy/hr by a 3 fraction HDR implant. What dose/fraction should be used to keep the effect on the tumour the same, assuming $\alpha/\beta = 15$ and $\mu = 1.4$ hr^{-1}?

1. 10.25 Gy
2. 11.32 Gy
3. 11.65
4. 10.92

MCQ Example

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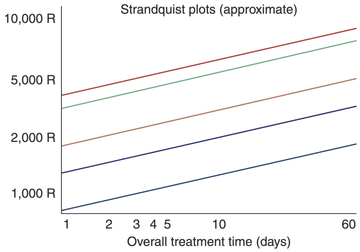
3. **11.65**

Table of Contents

Nominal Standard Dose (NSD)

- An empiric equation based on the Strandquist Plots:
  - Back in the 1930s-1940s, Strandquist treated a bunch of skin cancers and plotted radium exposure versus skin erythema, desquamation, necrosis, and tumor cure.
  - He found that the relationship between total dose and overall treatment time formed a series of parallel lines on a logarithmic chart.
  - The Strandquist plots (approximate). These parallel lines represent various skin-related endpoints such as erythema, desquamation, and necrosis.



Nominal Standard Dose (NSD)

Continue

- Unlike newer models, the Ellis NSD has no theoretical basis whatsoever; it is an empiric "curve fitting" model.
- The NSD can accurately predict acute skin toxicity and skin cancer response because that is where the data came from.
- The NSD makes no attempt to predict late toxicity.
- There are two versions of the equation: one with time and fractionation, another with just fractionation.
- For N fractions of total dose D delivered over T days:

$$D = \text{NSD} \times N^{0.24} \times T^{0.11}$$

Table of Contents

Cumulative Radiation Effect (CRE)

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Table of Contents